

Mining Attention Dynamics: Auditing KV-Saliency Prediction in LLMs

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Abstract—Selecting useful key–value (KV) state requires evaluating predictions under the budget and candidate set of each retention decision. KVSaliencyBench audits these evaluation objects through future-attention labels, explicit selector semantics, request and source-prompt splits, and versioned collection and validation code. On corrected traces from two instruction-tuned models, a three-parameter logistic scorer achieves pooled area under the receiver operating characteristic curve (AUC) of 0.890, close to LightGBM’s 0.896. Yet the within-layer attention signal recovers 0.659 of each decision’s future top-decile blocks, versus 0.613 and 0.615 for those models. This ordering difference narrows at 20% retention, showing why pooled ranking and decision-level budgets must be evaluated separately. A separate, pinned bfloat16 masked-loop rerun over 14 model–dataset cells finds the runtime-reconstructed scorer and an accumulated-attention baseline equivalent in mean LongBench F1 within a specified tolerance of two absolute points; both remain below full-cache quality. Historical served-oracle diagnostics favor the baseline, but use a different target and experiment version; we audit capacity-forced misses using logged counts. A feature bridge measures reconstruction effects without identifying the cause of the historical proxy gap. Reference backends reduce physical KV storage by about 5.3 times at 20% retention, with much smaller reductions in total allocated memory. Extended mask/physical tests produce logit discrepancies that exceed the preset numerical tolerance. These bounded findings support evaluation alignment, not a production serving or universal policy-equivalence claim.

Index Terms—KV cache, attention prediction, empirical audit, calibration, benchmarking, closed-loop evaluation.

I. INTRODUCTION

KV state grows with context while attention concentrates on a subset of positions. KV-management methods exploit this through accumulated attention [1], recent-query observations [2], query–key page bounds [3], cross-layer information [4] and learned retention [5]. Better prediction need not identify a better retention policy: selection changes later model states and their outputs. Traceable evaluation of inference state under a budget connects this study to D2AI’s infrastructure, provenance and benchmark scope.

We audit which signals predict future attention, whether predictor complexity helps, and how proxy rankings relate to answer quality. Offline labels, served-oracle diagnostics and closed-loop answer scores are distinct objects; the runtime scorer also reconstructs its features.

Contributions. (1) *A protocol and benchmark.* KVSaliencyBench fixes future-attention labels, selector semantics (Table I), request- and source-prompt splits, tie-aware per-decision metrics and versioned collection. (2) *Controlled findings.* Two attention-magnitude views support a three-parameter logistic scorer within 0.006 AUC of LightGBM on v2 traces without a calibration step. At the per-decision top-decile budget, the within-layer signal recovers more future top-decile blocks than either; this reversal shrinks at 20% retention. A separate pinned bfloat16 masked-loop rerun finds the reconstructed scorer and accumulator mean-F1 equivalent within a pre-specified ± 0.02 margin over 14 model–dataset cells (TOST $p = 0.0007$). A bridge measures feature reconstruction (§IV-B). (3) *An audit and its repair.* We diagnose corrupted float16 Qwen2.5 generation, rerun with a pinned simulator, audit capacity effects from logged counts, and report physical-KV parity levels. Code, tests and provenance accompany the paper.

II. PROBLEM AND EVIDENCE LEVELS

A. Attention-defined saliency

At layer ℓ and decode step t , let $\mathcal{B}_{\ell,t}$ contain the logical KV blocks currently under consideration. A block groups contiguous original token positions. We aggregate attention by averaging over heads and over a block’s token positions; denote the resulting statistic by $a_{\ell,b,t}$. It is a block-mean attention statistic, not necessarily a mass normalized to sum to one over blocks. For a horizon h and a fraction r , the target is $z_{\ell,b,t}^{(h)} = \mathbf{1}[b \in \text{Top}_{\lceil r|\mathcal{B}_{\ell,t}| \rceil} \{a_{\ell,b',t+h} : b' \in \mathcal{B}_{\ell,t}\}]$; ranking is restricted to blocks present at t . We use $r = 0.1$ and horizons $h \in \{1, 4, 16, 64\}$ (the archived collector’s terminal-horizon caveat is in §IX).

Versioned inputs / recorded provenance

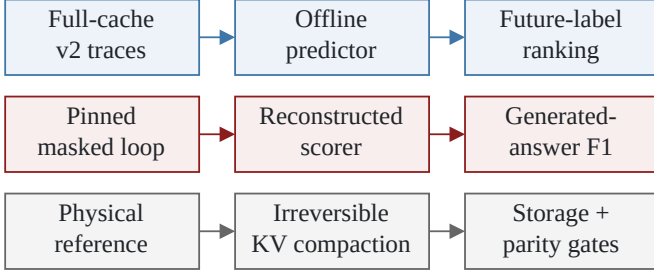


Fig. 1. Evaluation paths. Runtime features and selector semantics differ across paths (Table I); physical-reference parity is tested separately.

A scorer $f(\mathbf{x}_{\ell,b,t})$ estimates the label probability from recorded features; examples arrive in decode order, positives are a small fraction of blocks, drift is a hypothesis to test, and serial dependence means millions of block rows are not millions of independent observations.

B. Three different evaluation targets

Fig. 1 locates the evaluation paths. Offline prediction scores future-attention labels on fixed full-cache traces; served-oracle diagnostics score a current-attention target along the evaluated trajectory, and attention importance is not answer correctness; closed-loop task quality scores answers the selector’s own decisions helped produce, here through masking rather than a cache manager. A fraction of blocks allowed to attend is a *logical retention budget*, not a fraction of device memory released.

C. Statistical protocol

Offline ranking uses AUC, threshold-based AUPRC (tied scores form one group, so the value cannot depend on row order), precision/recall at $k = r$, Brier score and expected calibration error (ECE). Top- k metrics are reported pooled over test rows *and* per cache decision (request, layer, step) at the label’s own $\lceil rn \rceil$. Training and testing use disjoint requests and paired bootstrap intervals resample requests, not block rows. Unless marked otherwise, intervals are 95%; equivalence intervals are 90%. A 10,000-draw sensitivity analysis jointly resamples all held-out model-requests sharing a source prompt, on both the original request split and the source-disjoint split. Top- k score ties use a stable descending sort: pooled ties follow seeded sample row order, and within-decision ties follow increasing block index. V2 label and cross-layer indicator boundaries retain the collector’s NumPy `argpartition` tie rule (implementation-dependent on the original block order), with exactly $\lceil rn \rceil$ positives; these are not randomized ties. The main downstream comparison aggregates paired request scores into architecture–dataset cells and a TOST analysis [6], [7] asks whether their mean difference lies inside a pre-specified margin. Failure to detect a difference is not equivalence, and equivalence of two comparisons with the same baseline does not imply equivalence of the remaining pair.

III. SIGNALS AND EMPIRICAL FEATURE SELECTION

A. Recorded views

The feature family contains a normalized within-layer attention EMA $x_{wl} = \bar{a}_{\ell,b,t} / \max_{b'} \bar{a}_{\ell,b',t}$, a previous-layer hot-block indicator $x_{cl} = \mathbf{1}[\bar{a}_{\ell-1,b,t} \text{ is in the top-}r]$, a query-derived proxy $x_q = (1 + \cos(\tilde{q}_t, K_{\ell,b}))/2$ and a temporal-age feature $x_{age} = \exp(-(t - p_b)/w)$. Here $\bar{a}$ is an EMA and p_b the block-creation step supplied by the collector, so x_{age} is an age proxy, not an LRU access-recency measurement; the tildes mark the archived query representation, whose coordinate-space audit in §IX limits conclusions about that view. The cross-layer signal is an attention-magnitude carry-over, not InfiniGen’s query-based mechanism; the offline extractor uses the current layer’s own EMA as its first-layer fallback, whereas runtime realizations need their own explicit fallback and alignment rules (Table I).

B. Relevance and interaction diagnostics

We estimate relevance $I(X_i; Z)$ and interaction information

$$\Pi(X_i; X_j; Z) = I(X_i; X_j | Z) - I(X_i; X_j)$$

with quantile-binned plug-in estimators. On the version-2 corpus, within-layer EMA and the cross-layer indicator have $I(X_i; Z)$ of 0.104 and 0.099, versus 0.007 for the cosine query proxy and under 0.001 for age (Fig. 2); raw-probe AUCs are in Table II. We omit the historical single-view fitted/raw AUC comparison because its sample and aggregation correspondence cannot be certified: a positive monotone fit alone preserves AUC. The signs of small interaction estimates are *descriptive*: small pairwise interaction implies neither conditional independence nor the absence of higher-order effects.

Archived conditional-information diagnostics $U_i = I(X_i; Z | X_{-i})$, explored across binning grids and label permutations, are larger for the magnitude views than for the remaining proxies (≤ 0.006): within-layer is 0.133–0.155 on every grid, and the binary cross-layer estimate is 0.048–0.084 on the finer grids but collapses to zero under the coarsest one, so we report its grid sensitivity rather than the most favorable cell. These estimates motivate a candidate subset, evaluated on held-out requests; they are not a sufficiency certificate, and the evidence for the compact model is its measured predictive performance.

IV. PREDICTOR FAMILY AND FEATURE REALIZATIONS

A. A compact offline scorer

We fit a ridge-regularized logistic model by Newton–IRLS, $f(\mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b)$; on the within-layer and cross-layer features it has two coefficients and one intercept. Comparators only: a four-view logistic fit (“closed form” in the code for historical reasons, the same iterative fit), a 15-parameter interaction model and a 148-parameter tiny MLP. A proper log-loss fit without class reweighting has low measured ECE here and the class-balanced LightGBM does not, but an unweighted or recalibrated tree removes that difference (§V), so calibration is separate from ranking.

B. Offline predictor versus runtime reconstruction

The masking-loop scorer reconstructs the cross-layer view from the previous layer’s EMA, neutralizes the query view, and feeds selection into later states. Its task F1 therefore cannot inherit the offline predictor’s AUC. Table I distinguishes fixed offline features, policy-dependent runtime features and the irreversible physical reference. Feature, target and trajectory changes remain competing explanations of the archived offline-to-loop gap. **Bridge experiment.** With the frozen two-view checkpoint, one label (top- $\lceil 0.1n \rceil$ of block attention four steps ahead, among blocks present now and still present then) and the 128 frozen prompts of §VII, we score (A) offline-style features re-derived from the full-cache block-attention stream under version-2 conventions, (B) the reference backend’s runtime realization on the same rows, and (C) that realization along the on-policy trajectory at 20% budget, whose candidates are the retained blocks. Mean held-out request AUC is 0.902/0.899/0.989 on Llama-3.1-8B (bfloat16) and 0.894/0.892/0.989 on Qwen2.5-7B (float32): the feature realization costs -0.003 $[-0.004, -0.001]$ (84/128 requests negative) and -0.002 $[-0.003, -0.000]$ (75/128), whereas the on-policy evaluation yields higher AUC by $+0.090$ and $+0.098$ (128/128 both), but jointly changes the trajectory and candidate set; it does not isolate a feedback effect. Neither contrast reproduces the archived drop from 0.948 to 0.643. The bridge also differs from that archive in corpus, aggregation and numerics. Attribution would require a label-only arm on identical trajectories, candidates, features and scorer; without it the source of the archived gap remains open.

C. Selector semantics across the three evidence objects

Table I lists the rules that decide what each object measures. Two differ on purpose: the archived loop recomputes its kept set every eight steps, so an excluded block can return, and budgets a rounded fraction of all tokens including generated ones, whereas the reference deletes irreversibly and fixes the budget from the prompt. Its rules are read from the pinned rerun revision; the version-1 producer is not pinned.

D. Thresholds and online updates

A score can drive a fixed-size top- k set or a thresholded set $S_{\ell,t} = \{b : f(\mathbf{x}_{\ell,b,t}) \geq \tau_\ell\}$. The former fixes logical retention, whereas the latter lets set size vary. We evaluate static and online variants separately. Threshold calibration is described in §VI; its empirical in-distribution performance is not a hard per-request risk guarantee.

V. PREDICTION MEASUREMENTS

Data and comparators. Corrected version-2 traces carry the main offline evidence; historical extensions are labeled separately. Single-signal rows score one recorded feature and are not reimplementations of the papers that motivate them; learned comparators are LightGBM with 150 trees (a tree count, not a parameter count), a fitted MLP, a one-pass online logistic baseline and the compact family.

TABLE I
SELECTOR AND LABEL SEMANTICS. BLOCKS ARE 32 TOKENS; n IS THE CANDIDATE COUNT; L IS PROMPT LENGTH; ORACLE SIZE K IS DEFINED IN EXP#7.

rule	v2 collector	masked loop	physical reference
attention	head/token mean; ragged block over members	head-group/query mean; ragged block padded	head/query/token mean
EMA	decay 0.9 from first decode value	H2O: last-32 sum; learned: last-8 EMA, 0.9	decay 0.9; last-64 prefill queries
within view	$\bar{a}/(\max \bar{a} + 10^{-9})$	$\bar{a}/(\max \bar{a} + 10^{-9})$	$\bar{a}/\max \bar{a}$
cross view	previous-layer EMA top- $\lceil 0.1n \rceil$; layer 0: own EMA	same rank rule; layer 0: own EMA	same rank rule; layer 0: zero
query / age	post-RoPE q ; age from block creation	query 0.5; age since last top- k	unused
label	future top- $\lceil 0.1n \rceil$ among current blocks; real lookahead	current unmasked top- K (Exp#7)	unused
budget	—	round($0.2n$), every 8 steps	$\lfloor 0.2 \lceil L/32 \rceil \rfloor$, prompt-fixed
mandatory	—	learned: 4 sink + 4 recent; H2O: half heavy hitters, half recent	1 sink + 1 recent
re-admission	—	allowed at each decision	irreversible deletion
new tokens	feature rows	unmasked until next decision; budget grows	eligible; fixed budget

A. Exp#1: Ranking and calibration

Version-2 corpus. Table II uses checksum-verified traces: phase-aligned queries, the prefill-predicted first token, genuine 64-step lookahead labels, 32 feature steps, and the same 128 LongBench QA prompts (64 Qasper, 64 NarrativeQA) for Llama-3.1-8B and Qwen2.5-7B, giving 30.4M rows and 256 model-requests. We sample 120K training rows and 150K held-out rows for pooled metrics; per-decision metrics use every held-out row, and AUPRC is threshold-based. A source-prompt-level split, holding out the same 32 prompts for both models, agrees with the request-level AUCs to three decimals for the two-view fit and LightGBM and within 0.006 for every method; this split control shows no substantial change, not proof of independence. The two-view fit reaches AUC 0.890 (source-prompt bootstrap $[0.882, 0.898]$) against 0.896 for LightGBM, a paired deficit of -0.0057 $[-0.0068, -0.0046]$, larger than the archived -0.0010 but still small; the archived two-view weights, applied frozen to the version-2 features, achieve the same rounded AUC (0.890), with different top- k values (Table II), and are slightly less calibrated (ECE 0.019 versus 0.006). The four-view fit is worse (0.873): dropping its cosine query proxy *raises* AUC by 0.012, and the phase-aligned dot-max probe adds 0.001 to the two views (0.891), a pooled near-zero that averages over opposite signs in the tested configurations (§IX). At the per-decision operating point the ordering changes: the raw within-layer EMA recovers 0.659 of each decision’s future top-decile blocks against 0.613

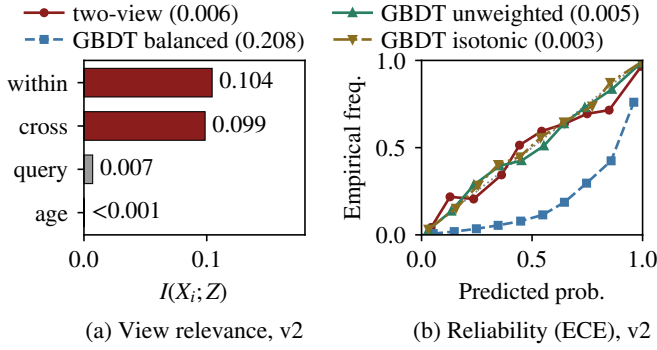


Fig. 2. Exp#1 on the version-2 corpus. (a) Per-view relevance on the 400K-row MI sample. (b) Reliability on the held-out sample for the two-view logistic fit, the class-balanced LightGBM control, an unweighted LightGBM and an isotonic-recalibrated one; ECE in parentheses.

for the two-view fit and 0.615 for LightGBM, though its pooled top-decile precision (0.542) and AUC (0.873) are lower. Paired over the 64 held-out requests the gaps are $+0.046$ $[0.043, 0.049]$ and $+0.044$ $[0.041, 0.047]$, positive on every request. These intervals use 10,000 joint draws of 57 source prompts. On the source-disjoint split (32 prompts, both models held out), the respective gaps are $+0.047$ $[0.044, 0.050]$ and $+0.045$ $[0.043, 0.048]$. The reversal belongs to the operating point, not the scorers: at the 20% the closed loop budgets, per-decision recall is 0.840, 0.838 and 0.830 and the gaps fall to $+0.002$ $[0.002, 0.003]$ (52/64 requests) and $+0.010$ $[0.009, 0.011]$. A pooled ranking spends the budget unevenly across decisions and rewards cross-decision calibration a per-step budget never uses; the cost of that depends on the budget, which is why we report the operating point a policy imposes. The smaller gap at 20% is consistent with, but does not explain, Exp#8’s downstream mean-F1 equivalence. Relevance and reliability are in Fig. 2: the magnitude views carry $I(X_i; Z)$ of 0.104 and 0.099, the query proxies 0.007 (cosine) and 0.025 (dot-max), the age proxy under 0.001; the class-balanced LightGBM has ECE 0.208 while the unweighted (0.005) and isotonic (0.003) variants match the two-view fit (0.006) on identical test rows. For the tree calibration controls, 48 of the 192 training model-requests are reserved for isotonic fitting; the base trees use the other 144. The two-view fit uses all 192 training requests; the 64 test requests are excluded from both fitting stages. This is request-level calibration separation, not source-level independence within the training set.

Historical context. The unpinned v1 corpus contains 43,931,784 rows from four instruction-tuned families, 32 chat prompts each, with positive rate 0.105; 32 requests are held out. Its two-view AUC is 0.948 $[0.942, 0.954]$ against LightGBM’s 0.949 (paired deficit -0.0010 $[-0.0016, -0.0003]$). The full historical tables, calibration and repair details remain in the artifact; they are not additional v2 evidence.

B. Exp#2: Workload extensions and transfer

The archived workload extension (chat, multi-/single-document QA, code completion, summarization; English and

TABLE II
Exp#1: VERSION-2 RANKING AND CALIBRATION, HORIZON 4, REQUEST SPLIT. THE DEC. COLUMN IS TOP-DECILE PRECISION MACRO-AVERAGED OVER EVERY CACHE DECISION, BESIDE THE POOLED COLUMN. AP IS TIE-AWARE AUPRC; HISTORICAL TABLES REMAIN IN THE ARTIFACT (SEC. IX).

method	AUC	AP	P@k pool	P@k dec.	ECE
age proxy	.502	.102	.102	.208	.691
query proxy (cosine)	.599	.135	.147	.415	.258
prev-layer indicator	.783	.412	.609	.610	.080
within-layer EMA	.873	.609	.542	.659	.087
LightGBM, balanced	.896	.674	.626	.615	.208
fitted MLP	.874	.658	.621	.613	.005
two-view logistic, refit	.890	.667	.624	.613	.006
two-view, archived weights	.890	.667	.622	.611	.019
two-view + dot-max	.891	.667	.624	.613	.005
four-view logistic	.873	.657	.621	.613	.005

Chinese; 128 prompts per workload from each of the four families) keeps both magnitude views useful (single-view AUC about 0.88–0.90 and 0.81–0.86), with pooled AUC 0.929 against 0.934 for LightGBM (AUPRC 0.777 versus 0.803 under the archived row-order tie rule; the original rows are unavailable, so tie-aware AP cannot be recomputed) and a gap of about 0.016 AUC on short multi-turn chat. A shared standardized four-weight predictor has mean cross-model AUC 0.920 against 0.921 within model (drop 0.0012), an offline transfer measurement only.

C. Exp#3: Temporal variation and threshold behavior

In the archived within-generation split (fit on steps 0–77, test on 102–127) a frozen scorer keeps AUC 0.948, about the same as an oracle refit, whereas the online IRLS update reaches 0.919 (evidence against that update, not against online learning); across forced topic switches, salient-set Jaccard falls to about 0.35 from 0.83 within a turn. At target miss 0.1 a fixed threshold of 0.5 gives second-half miss 0.274 at retention 0.095, an offline-calibrated split-conformal threshold 0.087 at retention 0.264 and the adaptive update 0.100 at the same retention: the calibrated static threshold carries the result, and no long-horizon trend is claimed from the old collector.

VI. AN EMPIRICAL RISK-TARGET CONSTRUCTION

Exp#4: Risk-target calibration. GuardKV maps a target miss rate to a per-layer threshold calibrated on held-out requests; in the archived in-distribution test, measured miss tracks the target within 0.007 and targets of 0.05 and 0.20 yield retention near 34% and 14%. The rule is explicit: with $s_{\ell,(1)} \leq \dots \leq s_{\ell,(n_\ell)}$ the sorted scores of layer ℓ ’s n_ℓ salient calibration rows, the threshold for target miss α is $\tau_\ell = s_{\ell,(k)}$, with $k = \min(n_\ell, \lfloor \alpha(n_\ell + 1) \rfloor)$ and the sentinel $s_{\ell,(0)} = 0$; an empty pooled positive set also gives threshold zero. Layers with under 64 salient rows use the pooled threshold, and the retained set is $\{b : p_b \geq \tau_\ell\}$ plus four sink and four most-recent blocks. This corrected implementation uses one-based ranks; the archived helper indexed the same rank from zero, one order statistic higher. The quoted historical results are unchanged empirical measurements, not verification of

the corrected rule. The split-conformal quantile has finite-sample marginal coverage only under exchangeability of calibration and test units, not established for correlated block/step rows, and marginal coverage would not imply a per-request bound; against the served-oracle target the reconstructed scorer needs about full retention for 90% coverage, and no out-of-distribution, hard-budget or production-risk guarantee is made.

VII. RUNTIME SCOPE AND SCORING COST

Masking is not physical eviction. The closed-loop experiments exclude selected blocks from attention and score generated answers; equal outputs under physical eviction would need matched token histories, kept sets, position semantics, feature availability and re-admission rules, none of which the simulator establishes, so its task-quality result is scoped to the masked loop and is not a serving claim (§IX).

Exp#5: Scoring-only cost. The archived microbenchmark times $28.7 \mu\text{s}$ at P99.9 for one 4096-vector batch of the four-weight kernel; it excludes feature extraction, selection and compaction and is not a serving TPOT.

Exp#6: Physical-KV implementation check. We evaluate an eager, single-request reference backend with original token positions, irreversible block eligibility and independent compact K/V storage, against a dense masked reference under the same eligibility rule (Table I). Gates compare native full-cache logits with the no-eviction path and replay a frozen teacher schedule between mask and compact storage. `h2o_block` is a shared-head block accumulator, not a reproduction of H2O or Ada-KV, and the runner implements no paging, offload, batching or production throughput.

Table III reports both models on one A100-80GB, Transformers 4.51.3, greedy decoding, the same 128 English QA prompts (64 Qasper, 64 NarrativeQA), input cap 4096, at most 128 new tokens and middle truncation on 104/128 prompts. Llama-3.1-8B-Instruct runs in bfloat16; Qwen2.5-7B bfloat16 parity exceeded the preset tolerance (maximum absolute error 0.25 versus 0.15) while argmax agreed, so Qwen is reported in float32. At 20% budget compact storage equals the live payload whereas the mask keeps full-size tensors, a $5.3\times$ ratio for both. Greedy sequences match between the two paths on 98/128 and 100/128 Llama requests (mean absolute F1 differences 0.014 and 0.007) and 128/128 Qwen requests. Allocator decode peaks fall only from 15.51 to 15.10 GiB (Llama) and 29.05 to 28.63 GiB (Qwen) because weights dominate, with prefill-plus-first-compaction peaks at 15.77 and 29.23 GiB. Two task clusters are too few for reliable cross-task equivalence inference; the 30% cells are in the artifact, and gates at the full input with 40 decode steps are in §IX.

VIII. PROXY RANKINGS AND CLOSED-LOOP TASK QUALITY

A. Exp#7: Offline and in-loop targets differ

The offline two-view AUC is 0.948 on the archived prediction task; on the served-oracle calibration set the runtime-reconstructed scorer reaches 0.643 (prompt-bootstrap 95%

TABLE III
EXP#6 (PHYSICAL-KV CHECK) AT 20% BLOCK BUDGET. STORED KV IS MEAN UNIQUE K/V TENSOR STORAGE (MiB) AFTER THE LAST STEP, LIVE KV THE MEAN RETAINED PAYLOAD, ITL (MS) THE MEDIAN LOCAL INTER-TOKEN LATENCY, NOT A REQUEST-LEVEL SERVING TPOT. LLAMA BFLOAT16, QWEN FLOAT32.

model	realization	F1	stored	live	ITL
Llama	full physical	.223	496	496	76.1
	H2O-style mask	.183	496	94	61.0
	H2O-style physical	.181	94	94	56.9
	reconstructed mask	.177	496	94	60.0
	reconstructed physical	.178	94	94	57.9
Qwen	full physical	.246	434	434	71.4
	H2O-style mask	.233	434	82	58.2
	H2O-style physical	.233	82	82	54.4
	reconstructed mask	.232	434	82	58.1
	reconstructed physical	.232	82	82	54.4

interval [0.632, 0.655], per-layer 0.54–0.76), but target, trajectory and feature realization change together (§IV-B). On four archived LongBench tasks at 20% retention (48 requests each), the accumulated-attention baseline has request-mean served-oracle miss 0.579 versus 0.623 for the reconstructed scorer (paired difference [0.033, 0.055], resolved on every task) while the per-task F1 intervals all overlap; the confirmatory comparison is Exp#8.

Capacity-forced miss and logged counts. For request j , step t and layer ℓ , the capacity floor is

$$F_{jtl} = \max(0, 1 - B_{jtl}^{\text{actual}}/K_{jtl}),$$

where B^{actual} includes mandatory blocks and K is the actual oracle-set size under the simulator’s minimum-32/top-decile rule, clipped to available blocks. The field `per_step_block_count` records $B_{jt}^{\log}$, the rounded layer mean of B^{actual} ; `per_step_B_t_oracle` records $K_{jt}^{\log}$, the rounded mean of populated layers’ K . We compute $F_{jt}^{\log} = \max(0, 1 - B_{jt}^{\log}/K_{jt}^{\log})$, then average steps within requests and requests equally (48 per task). Candidate count n is unlogged and unused. The pinned H2O-style and reconstructed policies fill a common B and form a common K across layers (IDs may differ), so scalar and layerwise floors coincide in that path. The artifact checks this invariant and documents block-boundary rules. The unpinned historical logs lack layer arrays: their reported floors are conditional recorded-count diagnostics, not independently verified layerwise floors. Uniform budget fractions alone would not establish equality. For the illustrative $n = 128$, $B = 26$, $K = 32$, the floor is 0.1875; the actual 4K-cap records have $B = 10\text{--}26$ and $K = 32$, giving mean floor 0.219 and residuals 0.360/0.404 from misses 0.579/0.623. All 192 paired request-mean logged-count floors match; recomputed residual intervals remain [0.033, 0.055]. Floors are 0.610 of miss 0.727 at $f = 0.10$, 0.022 at $f = 0.30$ and 0.004 at $f = 0.50$. Under the 16K cap, $n = 512$, $B = 102$, $K = 51$ would give zero, but recorded B spans 7–103 and K spans 32–51: the mean 0.063 comes entirely from $K = 32$, $B < 32$ steps in 87/448 requests with smaller

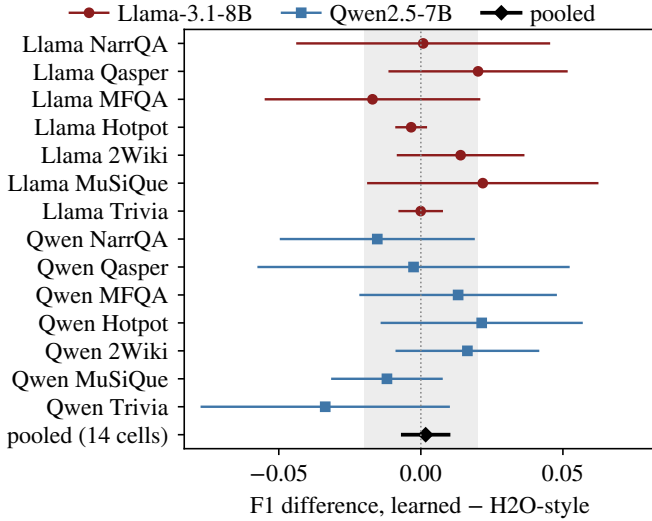


Fig. 3. Exp#8, pinned bfloat16 rerun (v2). Learned-minus-H2O-style F1 per architecture $\times$ dataset cell (64 paired requests, 95% t intervals) and the pooled 14-cell estimate (90% t interval); the band is the ± 0.02 margin.

effective budgets. Their paired floors are not identical, so residual intervals are recomputed in the artifact. Absolute miss includes capacity loss; policy differences remain comparable under matched budget rules, including Exp#9 at $f = 0.10$. That budget truncates the learned policy’s eight mandatory blocks on 3.3% of steps (minimum retained count: four). A scorer trained on the served-oracle target improves off-policy recall at 20% retention by 0.06 (interval $[0.03, 0.10]$) but worsens on-policy miss by 0.13 (interval $[0.10, 0.16]$), a reversal consistent with feedback-induced shift.

B. Exp#8: Matched-retention answer quality

The policy evaluated here is the runtime reconstruction of §IV-B, not the offline predictor of Table II. The main comparison uses chat-formatted LongBench QA [8], seven datasets, two architectures (Llama-3.1-8B and Qwen2.5-7B) and 896 paired prompts at 20% logical retention, each policy driving its own masked decoding trajectory.

Inferential unit and margin. The two architectures answer the same questions. Alongside paired TOST over 14 model-dataset cell means, we therefore average models within each dataset and test over seven dataset clusters (df 6). The pre-specified margin is two absolute F1 points, 5.4% of the rerun baseline 0.372: a practical tolerance, not a deployment threshold or preregistered one. Fig. 3 exposes task heterogeneity; mean equivalence does not imply equivalence on every task.

Pinned rerun. In bfloat16 no Qwen full-cache output contains a run of “!” (0/448 versus 181/448 archived) and 34/448 reach the 48-token cap (versus 210/448). Scored with the official LongBench scorer and all references, the learned scorer’s mean F1 difference from the accumulator over the same 896 paired prompts is +0.0017: the 14-cell TOST gives $p = 0.0007$ at ± 0.02 ; supplementary ± 0.01 tests give $p = 0.045$ (14 cells) and $p = 0.047$ (seven datasets), but that

TABLE IV
EXP#8 (MATCHED-RETENTION ANSWER QUALITY): MASKED-LOOP LONGBENCH F1 AT 20% RETENTION, 896 PAIRED PROMPTS, EVERY Δ AND TOST AGAINST H2O-STYLE ACCUMULATION. PINNED BFLOAT16 RERUN, OFFICIAL LONGBENCH SCORER, ALL REFERENCES.

policy realization	F1	Δ	$p \pm .02$
full cache	.419	+.047	—
H2O-style accumulation	.372	—	—
reconstructed two-view	.374	+.002	.0007
Ada-KV-style allocation	.363	-.008	.0080
PyramidKV-style allocation	.349	-.023	.582

margin was not pre-specified. Seven-dataset clustering gives $p = 0.0023$ at ± 0.02 . The 14-cell t-based and cluster-bootstrap 90% intervals are $[-0.0063, 0.0097]$ and $[-0.0056, 0.0087]$; the runner’s own scorer gives +0.0024 ($p = 0.0016$). Per architecture the difference is +0.0052 (Llama, $p = 0.016$) and -0.0018 (Qwen, $p = 0.026$). The Ada-KV-style comparator remains equivalent to the accumulator (-0.0084 , $p = 0.008$); the PyramidKV-style one does not (-0.023 , $p = 0.58$ at ± 0.02). Both allocation comparators use shared within-layer selection; these findings do not cover original H2O/Ada-KV implementations or imply equivalence between the learned and Ada-KV-style policies. This is equivalence between two compressed policies, not absence of compression loss: their F1 remains about 0.045–0.047 below full cache.

Historical audit. Archived Qwen float16 decoding emitted all-NaN logit rows, producing token 0 (“!”); a 56-prompt replay found this in 54 requests (687 NaN steps, 739 token-0 emissions), versus none in bfloat16/float32. The archived F1 difference was +0.00511 under a single-reference scorer (+0.0030 after official all-reference rescoring). The artifact retains the original scores, scorer checks, historical intervals and full repair record; Table IV reports the separate corrected rerun.

C. Exp#9: Budget and context checks

Sweeping retention over $\{0.10, 0.20, 0.30, 0.50\}$ yields no resolved per-budget quality gain, and the interval for the retention difference needed to reach 90% of full-cache F1 includes zero; neither proves curve equivalence or zero memory saving. At 16K the larger Llama sample (seven datasets, 448 prompts, 20% retention) has mean F1 difference -0.0006 , request-level 95% interval $[-0.012, 0.012]$ (exploratory TOST $p \approx 0.0009$, a different procedure from the 14-cell analysis); a 160-prompt pilot had suggested +0.010, which the expanded sample does not sustain, so we retract the context-growing win. Five alternative Llama prompt subsets give archived means between -0.008 and +0.011 and are not replications; a needle probe whose apparent advantages depended on repeated filler and lexical cues is a negative control, not proof that query-aware selection is unnecessary.

IX. LIMITATIONS AND ARTIFACT AUDIT

External validity. Historical traces cover four instruction-tuned 7–8B families; corrected v2 traces and the pinned task-

quality rerun cover two, at up to about 4K context. The archived 16K Llama and short-context Qwen2.5-14B runs are separate checks, not v2 replications (14B F1 0.301/0.307/0.301 for accumulator, reconstructed scorer and adaptive-allocation comparator); nothing is extrapolated to MoE, base checkpoints, 32K contexts or sampling, and decoding is greedy.

Query-coordinate audit. The legacy extractor hooks `q_proj` before rotary transformation but reads post-rotation keys, so it never computed the attention-space query-key product, and its revision is unpinned; we withdraw the “faithful Quest” reading of the old dot-max probe and claim no intrinsic query redundancy. Version 2 aligns queries and keys after RoPE, with a one-head test that its scaled dot products reproduce the model’s attention probabilities; even so a dot-max probe is not Quest’s page-bound algorithm. Adding it to the two magnitude views changes held-out *request-mean* AUC by -0.0070 on the 128 Llama prompts (32 test requests, all negative; $[-0.0082, -0.0058]$; two-view mean 0.899) and by $+0.0020$ on the matched Qwen prompts (31/32 positive; $[0.0016, 0.0024]$; mean 0.884): small, model-dependent changes with opposite signs in the two tested configurations, which the pooled version-2 value averages away. Some legacy long-context drivers also disabled prefill attention, depriving query-aware selectors of observations while leaving accumulators usable, so neither those comparisons nor the needle probe rank faithful query-aware implementations.

Generation and label boundaries. The legacy collector refeed the final prompt token to start decode and clamped unavailable horizons to the last recorded step; version 2 consumes the prefill-predicted token and generates genuine lookahead. No archived number was silently replaced, and the archived horizon curve is not evidence for long-horizon generalization.

Generation path and scorer of the archived loop. The archived cells came from an unpinned simulator in float16, whose Qwen2.5 path emits all-NaN logit rows (§VIII), scored against one reference with articles removed before punctuation. The rerun pins the revision by hash, uses bfloat16 for both architectures and applies one memory patch (a placeholder replaces each layer’s already-ingested attention weights in the tuple returned to the decoder layer; verified on a tiny model to change no statistic, mask or token). No rerun latency is reported; the GPUs were shared.

Tied scores and shared prompts. The archived AUPRC integrated one row at a time after a stable sort, crediting tied scores for row order; the two discrete probes are affected and cannot be recomputed, since the version-1 rows are gone, so Table II uses the threshold-based definition. The four archived families shared 32 prompts, so the archived split holds out model-requests, not source texts; version 2 adds a source-prompt split.

Inference and calibration. Block-level information estimates depend on binning and conditional sparsity; overlapping marginal intervals are not an equivalence test; the main TOST treats cell means as independent units, so its conclusions hold for this fixed collection of two architectures and seven

tasks, exploratory samples are not replications of it, and non-rejection does not establish no benefit.

Parity levels of the reference backend. The committed gates (1024 tokens, four decode steps) pass their numeric tolerance. Extending them to four prompts from both datasets, 1024–4096 input tokens and 40 decode steps with repeated eviction separates three levels: the session reproduces native full-cache logits exactly (maximum error 0 in all 12 runs); the compact and masked paths, fed the same teacher tokens and schedules, drift apart over successive evictions to a maximum logit difference of 0.66–1.44 in bfloat16 ($3.5\text{--}5.9\times$ the preset tolerance) and $3.6\times 10^{-5}\text{--}1.9\times 10^{-4}$ in float32 ($1.3\text{--}6.1\times$), so the extended test fails its preset tolerance. That the difference falls by four orders of magnitude with precision is consistent with floating-point ordering accumulated over 40 evictions, but these runs do not exclude a semantic difference; settling it needs a layer-by-layer comparison of retained indices, absolute positions and K/V values against a higher-precision reference. Top-1 agreement is 972/984 gated steps in bfloat16 and 984/984 in float32; free-running greedy parity is 98–100/128 (Llama) and 128/128 (Qwen). Tolerances were not raised; the records are in the artifact.

System boundary. The old physical-memory probe swallowed decode exceptions, put full prefill in its peak statistic and had no verified absolute-position or repeated-eviction protocol; it is retired, its source and results kept for provenance. Scoring microseconds, kept-set sizes and synthetic latency components measure none of production TPOT, HBM release, contention or throughput.

Availability. The artifact contains LaTeX, result JSON, analysis code, predictor weights, tests, the version-2 collector and the reference validator. The rerun’s exact simulator source is bundled with its MIT notice and verified against all 70 cells’ provenance hashes. A CPU example, environment and data/model preparation instructions accompany it; model weights remain separately licensed. Code, records and figure sources are at <https://github.com/lzbaclz/KVSaliencyBench>; raw traces are available by request.

X. RELATED WORK

Streaming learning studies evolving examples and distribution shift [9]; multi-view learning and relevance–redundancy analysis motivate comparing information sources [10], [11], which we apply to recorded attention dynamics without treating a small information estimate as a sufficiency proof. Calibration and conformal prediction [12]–[14] provide tools whose guarantees do not transfer to dependent runtime observations unchecked.

KV-management methods use heavy hitters, recent observations, cross-layer signals, query bounds, adaptive allocation and learned retention [1]–[5], [15], [16]; PagedAttention [17] addresses physical serving. Our archived feature probes and shared-selection allocation comparators are not interchangeable with those systems; the contribution is an audit of the links between intermediate prediction and task-quality measurements, not a claim that these methods are unnecessary.

Closest to our question are evaluation studies of KV compression. Gao et al. [18] audit compression from the serving side: unmeasured throughput costs, longer outputs, and per-sample quality variation hidden by averages. Agrawal and Mayer [19] benchmark task quality, throughput, time-to-first-token and realized compression ratio jointly. KVzap [20] is a recent learned pruning method evaluated on such a suite, and Locret [5] trains retaining heads on a causal-importance target and validates them by end-task quality. Task-level benchmarks such as LongBench and RULER [8], [21] score answers; KVSaliencyBench records intermediate features and attention-derived labels, a complementary level. Those works ask whether a method is worth deploying; we ask a narrower question they do not isolate, whether the saliency label, its feature realization and the feedback trajectory each preserve a proxy ranking, with controls that fix the scorer and the label while the feature realization and the candidate set change (§IV-B, Exp#7–8). The negative findings concern how proxies transfer to task quality in the tested implementations, not the value of learned or value-dependent selection [22].

XI. CONCLUSION

KV-saliency prediction and KV retention are related but distinct evaluation problems. On version-2 traces, the ranking advantage of learned scorers in pooled metrics reverses under per-decision top-decile budgets and narrows at 20% retention. A separate pinned bfloat16 masked-loop rerun finds the reconstructed scorer mean-F1 equivalent to an accumulator within ± 0.02 , while both lose quality relative to full cache. Historical served-oracle diagnostics favor the accumulator; they are not a same-version proxy–quality experiment. The bridge quantifies feature reconstruction at -0.003 AUC, but jointly changing trajectory and candidates does not explain the archived gap; a label-only arm is still missing. These findings concern the tested data and implementations, not the impossibility of better query-aware selection or a serving-efficiency result. Versioned collection, provenance and an explicit benchmark protocol are the artifact’s contribution to data and database systems for AI.

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REFERENCES

- [1] Z. Zhang, Y. Sheng, T. Zhou, T. Chen, L. Zheng, R. Cai, Z. Song, Y. Tian, C. Ré, C. Barrett, Z. Wang, and B. Chen, “H2O: Heavy-hitter oracle for efficient generative inference of large language models,” in *Proc. Advances in Neural Information Processing Systems*, 2023, pp. 34 661–34 710.
- [2] Y. Li, Y. Huang, B. Yang, B. Venkitesh, A. Locatelli, H. Ye, T. Cai, P. Lewis, and D. Chen, “SnapKV: LLM knows what you are looking for before generation,” in *Proc. Advances in Neural Information Processing Systems*, 2024, pp. 22 947–22 970.
- [3] J. Tang, Y. Zhao, K. Zhu, G. Xiao, B. Kasikci, and S. Han, “Quest: Query-aware sparsity for efficient long-context LLM inference,” in *Proc. 41st International Conference on Machine Learning*, vol. 235. PMLR, 2024, pp. 47 901–47 911.
- [4] W. Lee, J. Lee, J. Seo, and J. Sim, “InfiniGen: Efficient generative inference of large language models with dynamic KV cache management,” in *Proc. 18th USENIX Symposium on Operating Systems Design and Implementation (OSDI 24)*. USENIX Association, 2024, pp. 155–172.
- [5] Y. Huang, B. Yuan, X. Han, C. Xiao, and Z. Liu, “Locret: Enhancing eviction in long-context LLM inference with trained retaining heads on consumer-grade devices,” *Transactions on Machine Learning Research*, 2025.
- [6] D. J. Schuurmann, “A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability,” *Journal of Pharmacokinetics and Biopharmaceutics*, vol. 15, no. 6, pp. 657–680, 1987.
- [7] D. Lakens, “Equivalence tests: A practical primer for t tests, correlations, and meta-analyses,” *Social Psychological and Personality Science*, vol. 8, no. 4, pp. 355–362, 2017.
- [8] Y. Bai, X. Lv, J. Zhang, H. Lyu, J. Tang, Z. Huang, Z. Du, X. Liu, A. Zeng, L. Hou, Y. Dong, J. Tang, and J. Li, “LongBench: A bilingual, multitask benchmark for long context understanding,” in *Proc. 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024, pp. 3119–3137.
- [9] J. Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy, and A. Bouchachia, “A survey on concept drift adaptation,” *ACM Computing Surveys*, vol. 46, no. 4, pp. 1–37, 2014.
- [10] A. Blum and T. Mitchell, “Combining labeled and unlabeled data with co-training,” in *Proc. 11th Annual Conference on Computational Learning Theory*. ACM, 1998, pp. 92–100.
- [11] H. Peng, F. Long, and C. Ding, “Feature selection based on mutual information criteria of max-dependency, max-relevance, and min-redundancy,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 27, no. 8, pp. 1226–1238, 2005.
- [12] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proc. 34th International Conference on Machine Learning*, vol. 70. PMLR, 2017, pp. 1321–1330.
- [13] A. N. Angelopoulos and S. Bates, “Conformal prediction: A gentle introduction,” *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023.
- [14] I. Gibbs and E. J. Candès, “Adaptive conformal inference under distribution shift,” in *Proc. Advances in Neural Information Processing Systems*, 2021, pp. 1660–1672.
- [15] Y. Feng, J. Lv, Y. Cao, X. Xie, and S. K. Zhou, “Ada-KV: Optimizing KV cache eviction by adaptive budget allocation for efficient LLM inference,” in *Proc. Advances in Neural Information Processing Systems*, 2025, pp. 125 462–125 498.
- [16] Z. Cai, Y. Zhang, B. Gao, Y. Liu, T. Liu, K. Lu, W. Xiong, Y. Dong, B. Chang, J. Hu, and W. Xiao, “PyramidKV: Dynamic KV cache compression based on pyramidal information funneling,” in *Proc. Conference on Language Modeling*, 2025.
- [17] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. Gonzalez, H. Zhang, and I. Stoica, “Efficient memory management for large language model serving with PagedAttention,” in *Proc. 29th Symposium on Operating Systems Principles*. ACM, 2023, pp. 611–626.
- [18] W. Gao, X. Zhou, P. Sun, T. Zhang, and Y. Wen, “Rethinking key-value cache compression techniques for large language model serving,” in *Proc. Eighth Conference on Machine Learning and Systems*, 2025.
- [19] N. Agrawal and R. Mayer, “Benchmarking KV-cache optimizations across task quality and system performance for long-context serving [Experiment, Analysis & Benchmark],” Preprint, EPub Bayreuth, 2026.
- [20] S. Jégou and M. Jeblick, “KVzap: Fast, adaptive, and faithful KV cache pruning,” *arXiv preprint arXiv:2601.07891*, 2026.
- [21] C.-P. Hsieh, S. Sun, S. Krizan, S. Acharya, D. Rekesch, F. Jia, and B. Ginsburg, “RULER: What’s the real context size of your long-context language models?” in *Proc. Conference on Language Modeling*, 2024.
- [22] Z. Guo, H. Kamigaito, and T. Watanabe, “Attention score is not all you need for token importance indicator in KV cache reduction: Value also matters,” in *Proc. 2024 Conference on Empirical Methods in Natural Language Processing*, 2024, pp. 21 158–21 166.